Title: Predicting Vinho Verde Wine Quality Using Exploratory Data Analysis and Machine Learning

Name: Kitty Jenkins Phoenix

Student Number: 240740021

Word Count: 2534

Abstract: The Point of this project was to find the factors that can change the perceived quality of the vinho verde wines by using exploratory data analysis and machine learning. Originally, I analysed the data sets separately before combining them into one and analysing them further from there.

Most of the wines were around the middle range scores, with the white wines showing a slightly higher average quality than the reds. The content of alcohol percentage showed the strongest relationship with quality while volatile acidity showed a negative correlation.

Two models were used throughout the project: linear regression and logistic regression. Alcohol was found to be an important factor in predicting the quality of the wine but there wasn't a single variable that could fully explain the scoring. The models were also limited because the data did not include other useful information.

Linear Regression achieved an RMSE of 0.751, while Logistic Regression achieved 0.741 accuracy.

Table Of Contents

1. What Was Done And How

1.1 Dataset Overview

<u>The dataset was originally created by Cortez et al. (2009).</u>

The dataset that was used in this project contains two datasets that are related to the red and white variants of the Portuguese "Vinho Verde" wine. The number of instances are 1599 red wine samples and 4898 white wine samples. There are also 11 input variables and one output variable representing the quality of the wine. These datasets can be viewed as classification or regression tasks.

1.2 Data Preparation

Created a variable called alcohol_cat/alcoholCategory which separated the alcohol percentage into either low, medium or high. For the extension, made the variable isSweet which determined if the wine was sweet or not. After, split the data into training and testing sets.

1.3 Exploratory Analysis Of Wines

1.3.1 Red Wine Histogram

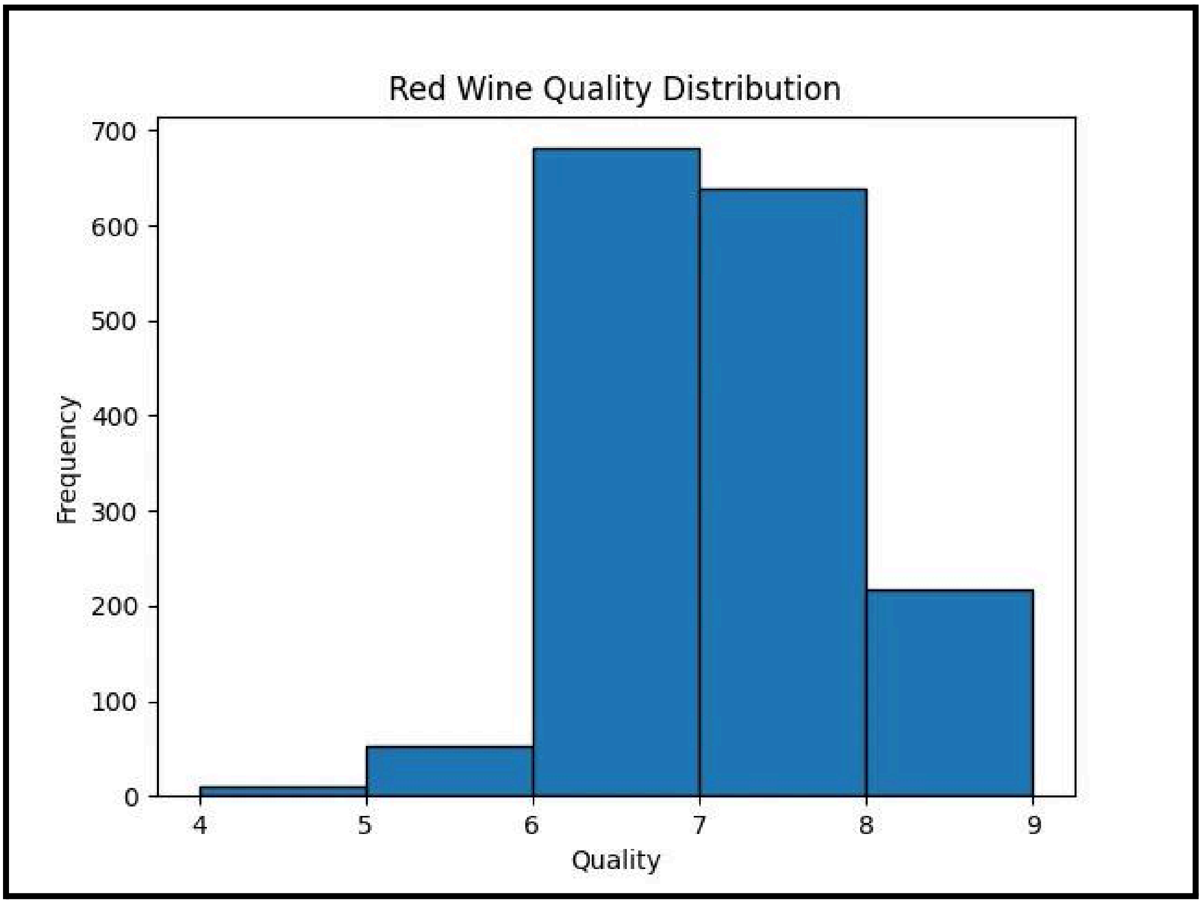

Figure 1: Red Wine Histogram

The most popular score was 6 with around 42.6 percent of wines being in that category. This means that wines are usually given an average quality score.

1.3.2 White Wine Histogram

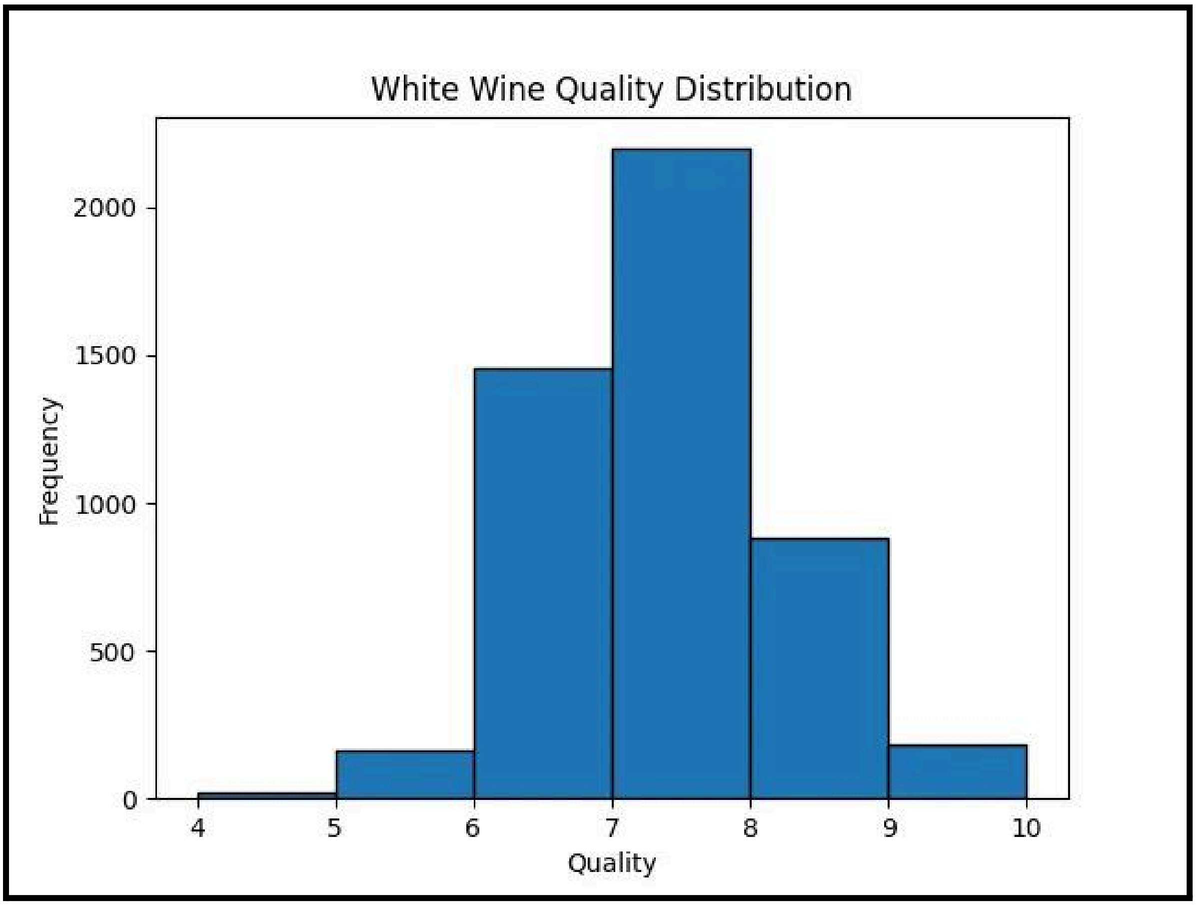

Figure 2: White Wine Histogram

The most common quality score is 7, with around 44.9% of wines.

1.3.3 Comparison

Most wines are concentrated around the middle quality scores, meaning very low and very high quality wines are less common.

1.4 Alcohol Categorisation And Analysis

1.4.1 Alcohol Content Categorisation

Alcohol content was grouped into three categories (low, medium, and high) using the mean and standard deviation of each dataset.

1.4.2 Quality Distribution Analysis

1.4.2.1 Red Wine Quality By AlcoholCategory

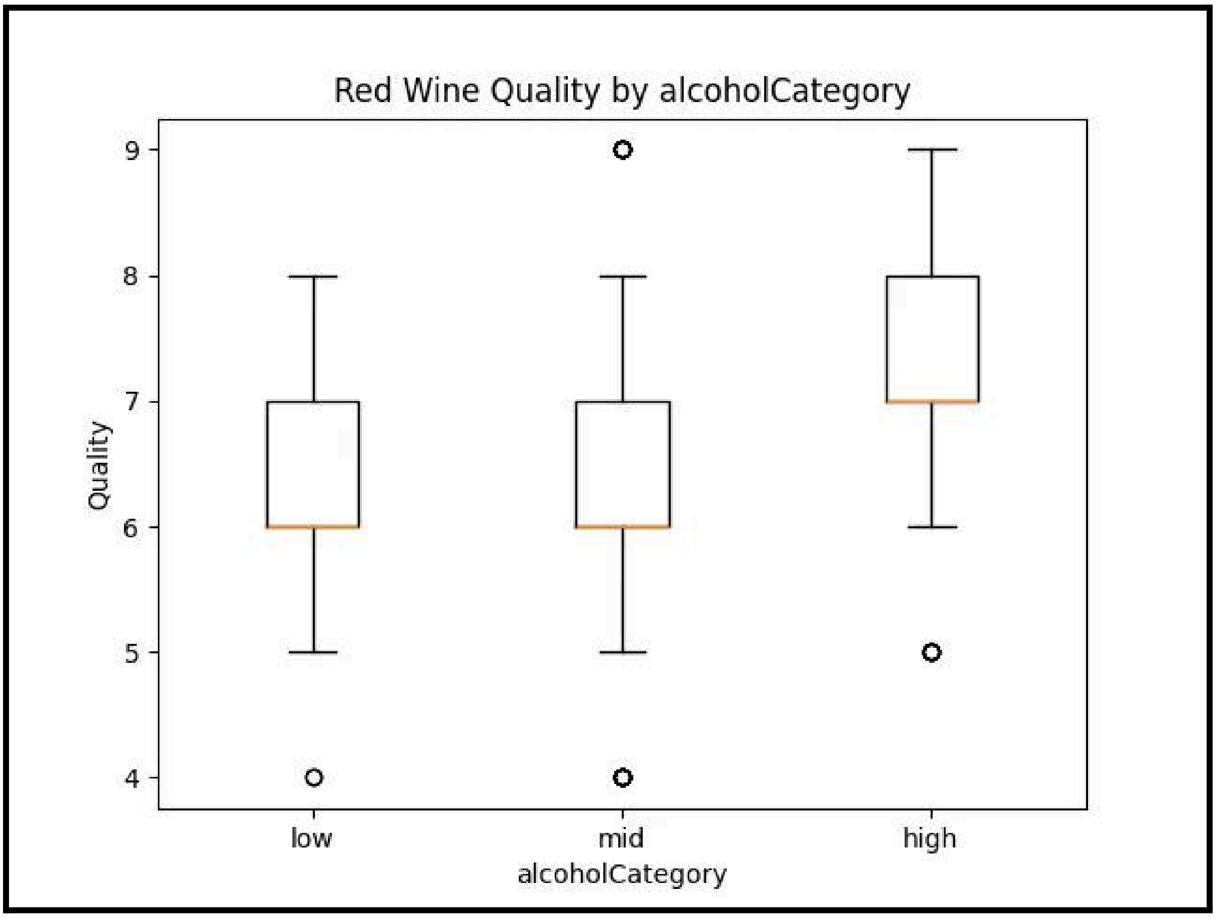

Figure 3: Red Wine Quality by alcoholCategory

1.4.2.2 White Wine Quality By AlcoholCategory

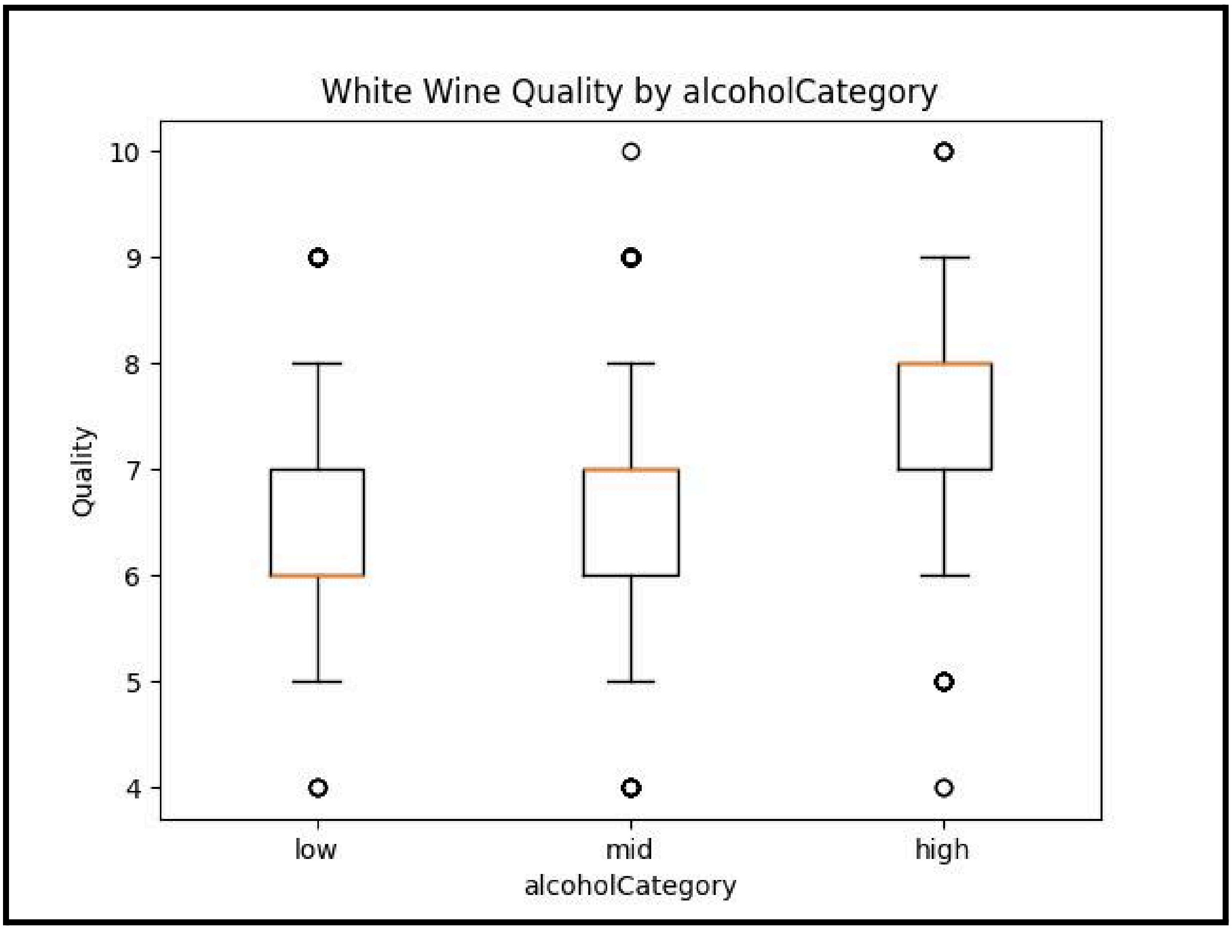

Figure 5: White Wine Quality by alcoholCategory

1.4.3 Conclusions

Higher alcohol categories consistently show higher average quality scores, indicating a positive relationship.

1.4.4 Visualisation Design

This boxplot is effective because it allows clear comparison between categories. It shows the median values, any outliers and the quartile ranges and it makes it easier to see the increasing trend in quality with higher alcohol levels.

1.5 Additional Feature Analysis (Extension)

1.5.1 Impact Of Wine Type

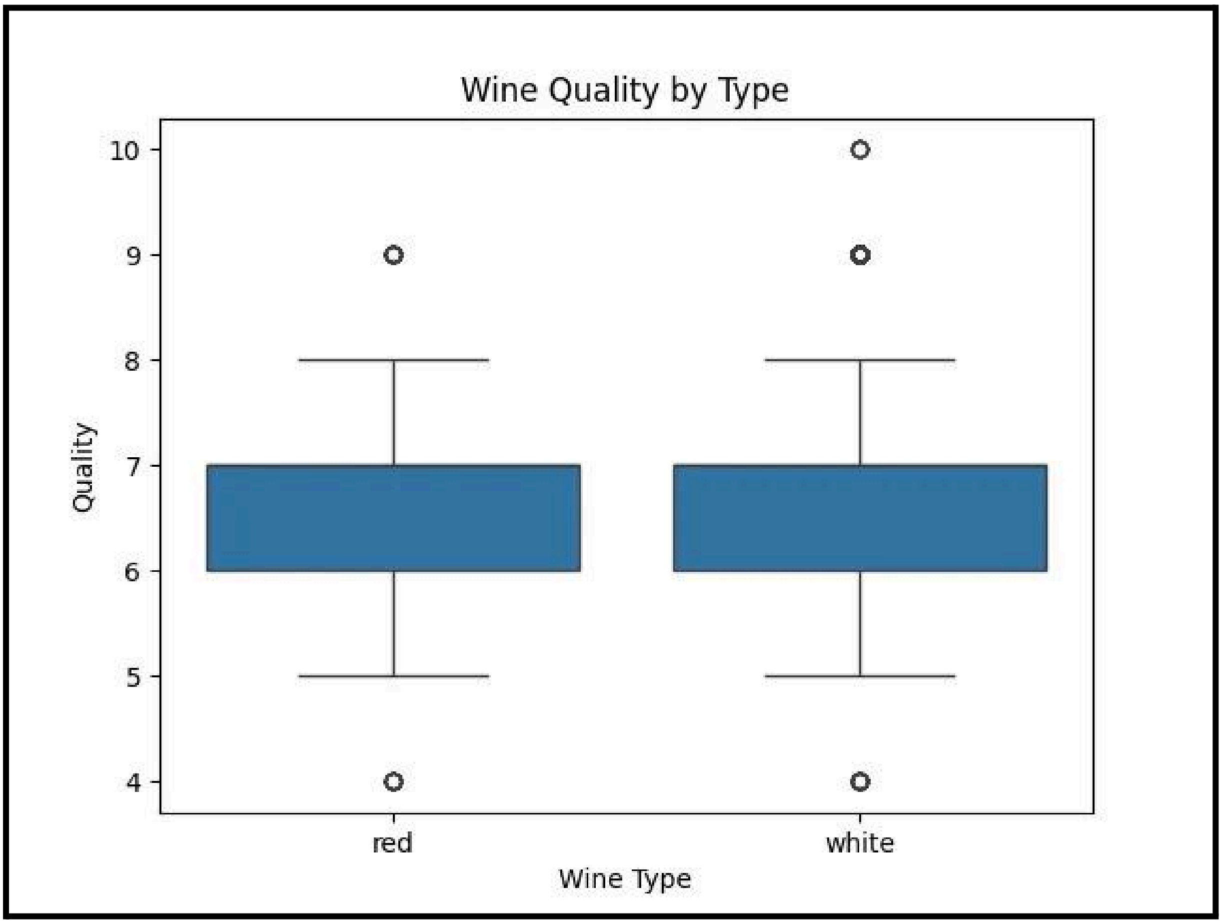

Figure 6: Wine Quality By Type

The boxplot shows that both red and white wines have a similar overall distribution of quality scores, with most values falling between 6 and 7. This suggests that the majority of wines in both datasets are of average quality.

1.5.2 Analysis of isSweet and Quality

The isSweet variable showed minimal impact on wine quality, suggesting residual sugar is not a strong predictor.

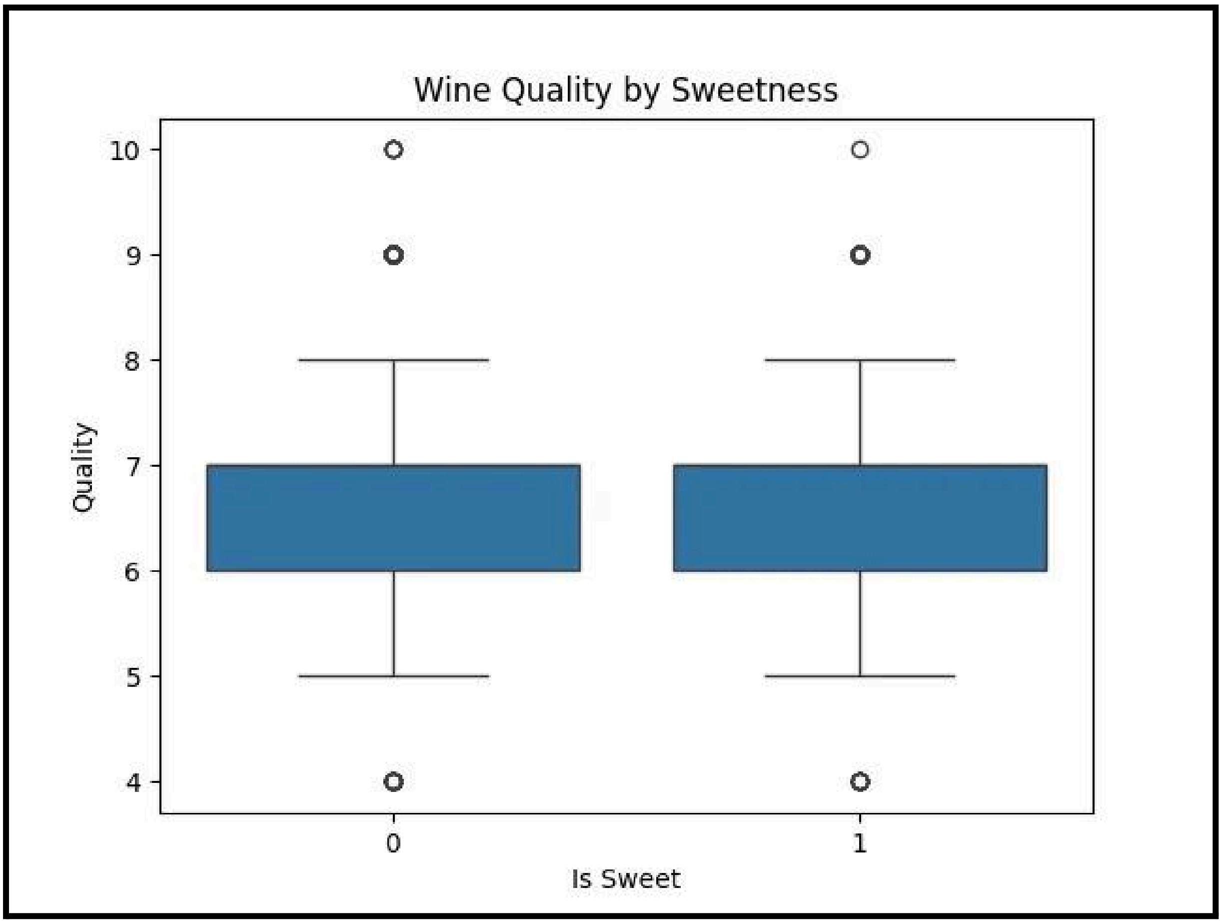

Figure 7: Wine Quality By Sweetness

This shows that residual sugar is not a strong prediction of the quality.

1.5.3 Combined Effects of Quality, alcoholCategory and isSweet

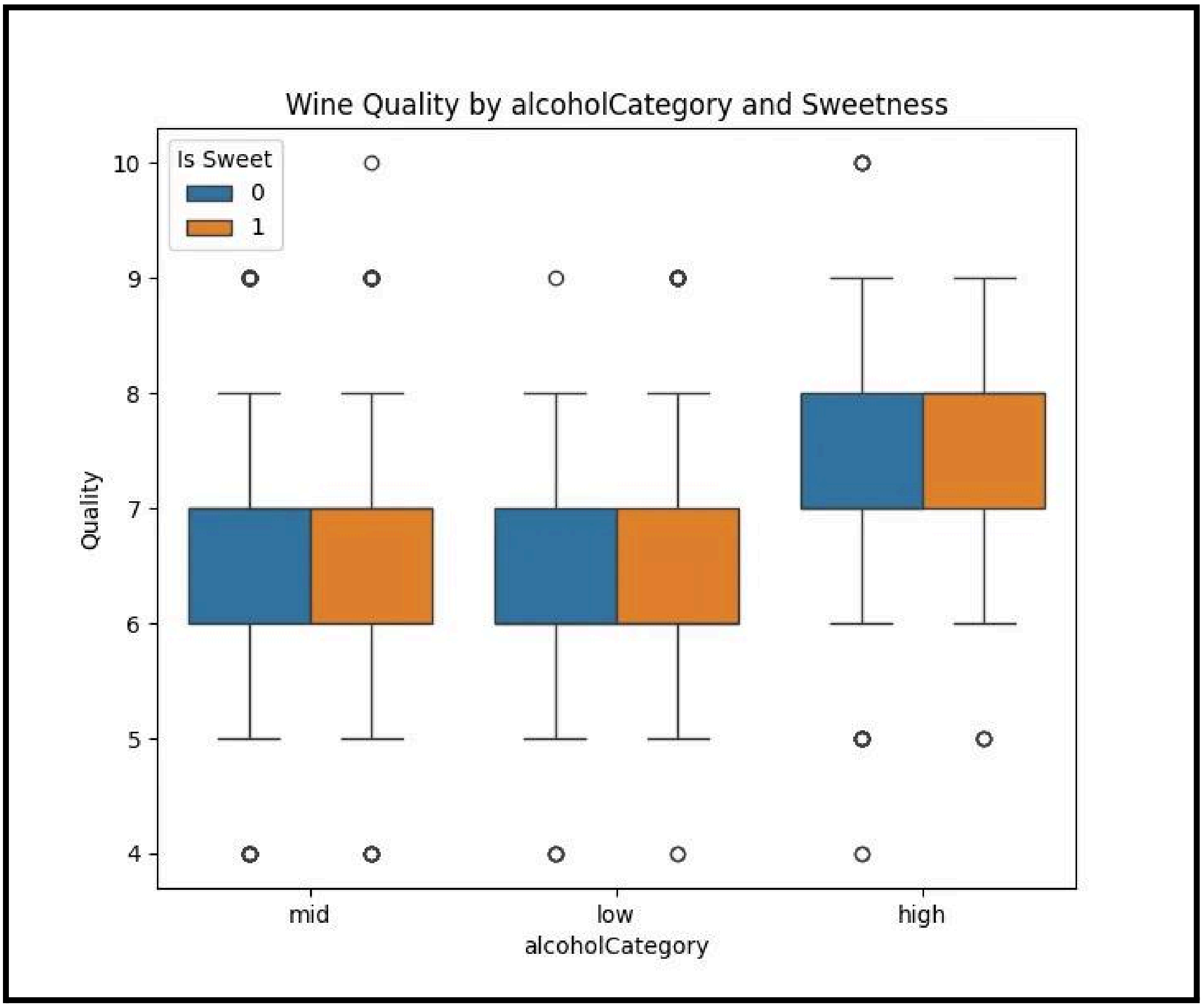

Figure 8: Wine Quality By AlcoholCategory and Sweetness

This graph shows that when the alcohol category is higher the quality is also higher in both data sets regardless of it being sweet or not and regardless of it being red or white.

1.6 Correlation Analysis And Variable Selection

1.6.1 Correlation Heatmap

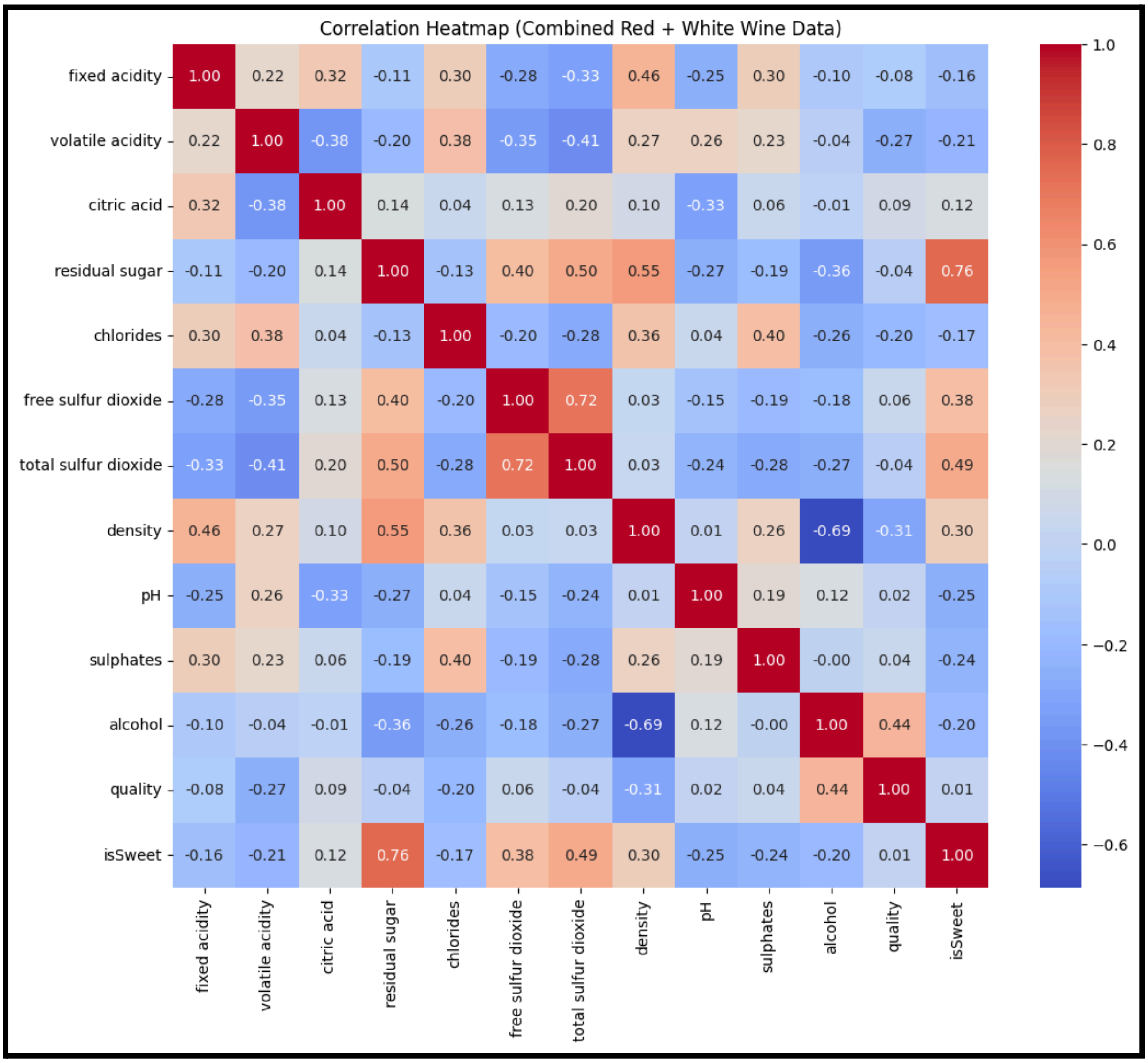

Figure 9: Correlation Heatmap (Combined Red + White Wine Data)

The heatmap uses colour to show the strength and direction of relationships, while position allows all variables to be compared at the same time. This makes it easier to identify strong predictors of quality and detect variables that are highly correlated with each other. Compared to individual plots, this approach provides a clearer overall view of the data.

1.6.2 Scatter Plot For Alcohol vs Quality By Wine Type

1.6.2.1 Non Interactive Chart

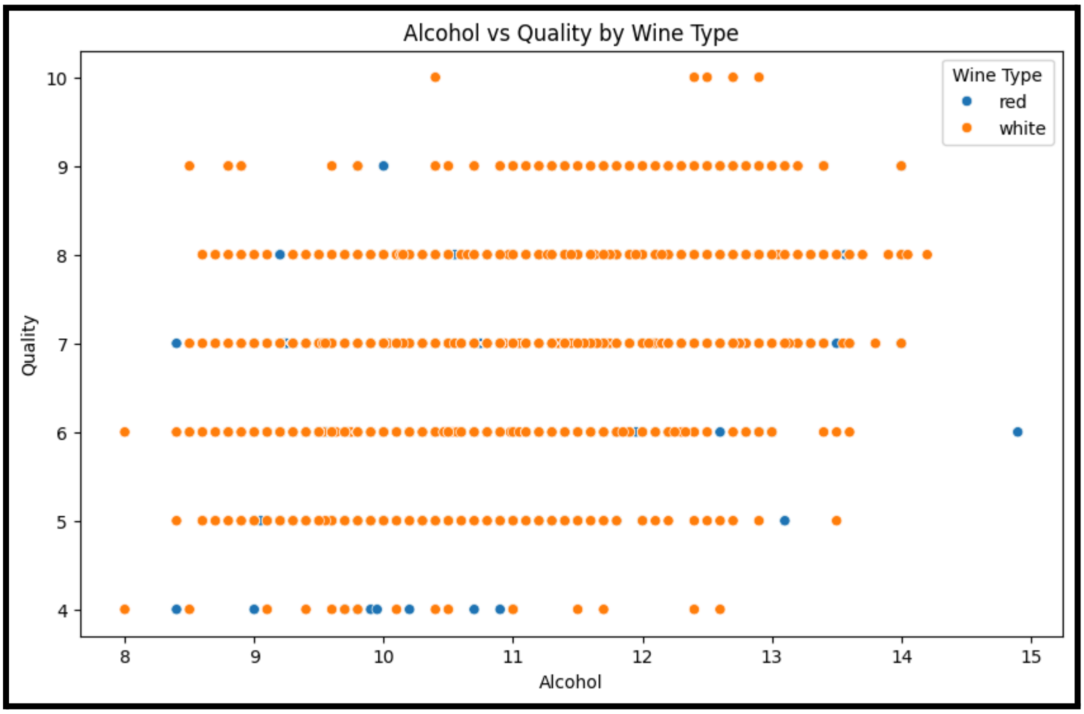

Figure 10: Alcohol vs Quality by Wine Type

1.6.2.2 Interactive Chart

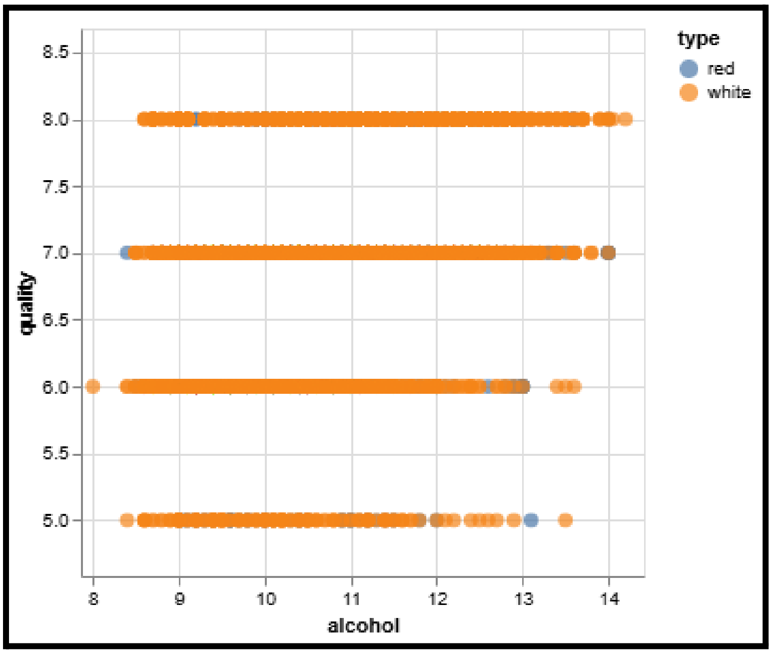

Figure 11: Interactive Chart

The scatter plot shows a positive correlation between the alcohol content and the quality of the wine. As the alcohol levels increase so does the quality around above 11 percent. However, there is still quite a lot of overlap between quality which means that alcohol alone does not fully determine wine quality.

1.6.3 Use Of Visualisation

Heatmaps show relationships between variables using colour and position. This makes it easier to spot strong positive or negative correlations and compare all variables in one view.This method is more effective because it allows all variables to be compared in a single view.

This suggests that some variables may not add new information to the model, which supports the need for selection of features.

1.7 Machine Learning Methods

Linear Regression and Logistic Regression are commonly used methods in data mining (Han et al., 2011).

1.7.1 Linear Regression

The linear regression predictions were made for both training and test datasets, and performance was evaluated using Root Mean Squared Error and Mean Absolute Error. This was done because it is simple and is very suitable for regression problems. However, Linear Regression assumes a linear relationship between variables, which may limit its ability to capture more complex patterns present in the data. This suggests that the relationship between variables may not be strictly linear.

1.7.2 Logistic Regression

For Logistic Regression, measured performance using accuracy and confusion matrices. Used this model to compare it with regression and to see if wine quality could be split into groups. Although, it assumes there is a linearity relationship between the dependent variable and the independent variables so it may not capture more complex patterns in the data.

1.7.3 Selection of Features Model (Extension)

The selection of features tests showed how changing the variables affected the model. When the wine type variable was removed, it did not make much difference to the results. When only the variables that were more strongly related to quality were used, the model still performed reasonably well. This shows that some variables are not that important and can be removed, especially when they are very similar to each other.

1.7.4 Additional Modelling Experiments (Extension)

An additional experiment was conducted to predict all quality labels instead of using binary classification. This showed that simplifying the problem into binary classification improves model performance, while predicting all labels provides a more complex but realistic task.

1.8 Method Of Evaluation

The performance of the models was evaluated by using training and test datasets. For Linear Regression, Root Mean Squared Error and Mean Absolute Error were used to measure the prediction of accuracy.

For Logistic Regression, the accuracy number was used for the evaluation. Performance was compared between training and test sets to detect any overfitting.

2. Results And Evaluation

2.1 Summary Of Exploratory Findings

The correlation heatmap supported this finding, as alcohol had one of the strongest positive correlations with quality. Volatile acidity showed a negative relationship with quality, suggesting that higher volatile acidity may reduce the wine quality.

White wines had a slightly higher average score than red wines (white had 6.88 and red had 6.64).

The variable with the best positive correlation with quality was definitely the alcohol percentage. In both types of wines, the high category had higher mean quality scores than the low and mid categories. Although, in the scatter plot it showed overlap between quality scores, meaning that the alcohol content alone cannot fully explain the quality of the wine.

2.2 Model Performance

Model	**Training Result**	**Test Result**	**Interpretation**
Linear Regression	RMSE = 0.720	RMSE = 0.751, MAE = 0.587	Good baseline model with similar train/test results
Logistic Regression, threshold ≥ 7	Accuracy = 0.746	Accuracy = 0.741	Best balanced classification threshold
Logistic Regression, threshold ≥ 6	Accuracy = 0.966	Accuracy = 0.956	Misleading because most wines are classed as high quality
Logistic Regression, threshold ≥ 8	Accuracy = 0.826	Accuracy = 0.812	Reasonable accuracy, but struggles with rare high-quality wines
Linear Regression without wine type	RMSE = 0.722	RMSE = 0.752	Very similar to the full model, so wine type added little value

Feature Selection Linear Regression	RMSE = 0.734	RMSE = 0.761	Slightly worse, but still reasonable
Logistic Regression, all labels	Accuracy = 0.553	Accuracy = 0.532	Much harder due to class imbalance

The Linear Regression model achieved a test RMSE of 0.751 and MAE of 0.587. The similar training and test RMSE values indicate that the model did not significantly overfit.

The Logistic Regression model using a threshold of quality ≥ 7 achieved a test accuracy of 0.741, providing a balanced classification between lower and higher quality wines. However, some misclassification occurred for wines with scores close to the threshold.

2.3 Comparison Of The Two Approaches

Linear Regression was useful because the error metrics showed how far predictions were from the true values and it was also easy to interpret, making it a strong machine learning model.

Logistic Regression was useful when the problem was simplified into binary classification. The threshold value of 7 was the best since it provided a middle ground and split the average and high quality wines quite evenly. The threshold of 6 outputted a very high accuracy score but it's a bit of a misleading value since most of the wines were put into the high quality group. A threshold of 8 was stricter, but there were fewer high quality wines, which made predictions harder.

Overall, Linear Regression was better for predicting the actual quality score, while Logistic Regression was better for answering a simple one or the other answer questions.

2.4 Limitations And Discussion

Most wines were rated around 6 or 7, while very low and very high scores were rare. This meant that it was harder for the models to learn any patterns for the higher or lower quality scores. This was noted when predicting all the different quality labels because it achieved an accuracy value of 0.53.

Another drawback was that the dataset only included physicochemical variables and not any other factors that could affect the quality of the wine (the grape type and price). This means the model is limited and misses important factors.

To improve the performance, more advanced models or a better selection of features could be used to capture more complex patterns in the data.

3. Conclusions

3.1 Summary

After analysing the data, it showed that most wines are given a quality score around the middle which means average quality wines are the most common in the dataset. Furthermore, the percentage of alcohol also showed that there is a strong positive correlation with the quality of the wine. Logistic Regression performed reasonably well for classification tasks while trying to predict all the quality labels was difficult due to having some more uncommon scores. The results from training and testing the models were pretty similar which shows us that the models work pretty well for unseen data.

Although, the performance was limited due to not having any additional contextual features such as the type of grape and price which could influence the overall quality of the wine.

3.2 Future Work

Future work could include adding more variables to improve accuracy. This could involve including external factors such as grape type, region, or price to give a more complete understanding of wine quality.

3.3 Personal Reflection

The project provided a lot of valuable experience in data analysis and machine learning and gave the stepping stones to begin some other data science projects. The biggest problem was trying to interpret all the different numbers that were being outputted and evaluating what they mean. Overall, the project has improved the technical and analytical skills and for any future work it would focus on improving understanding of evaluation metrics and exploring more advanced modelling techniques.

References

Cortez, P., Cerdeira, A., Almeida, F., Matos, T. and Reis, J. (2009) *Modeling wine preferences by data mining from physicochemical properties*. Available at: https://doi.org/10.1016/j.dss.2009.05.016

Han, J., Kamber, M. and Pei, J. (2011) *Data Mining: Concepts and Techniques*. 3rd edn. Burlington: Morgan Kaufmann.

Hastie, T., Tibshirani, R. and Friedman, J. (2009) *The Elements of Statistical Learning*. 2nd edn. New York: Springer. Available at: https://doi.org/10.1007/978-0-387-84858-7